

App Proposal: TrialBite

Project Essence

App Name: TrialBite

Mission Statement: A tool that tracks single-food trial tests step-by-step to help me systematically log, identify, and isolate allergic reactions.

Target Audience

While this is made specifically for me and my ongoing trials on my allergies, this could also be used by other individuals undergoing a food elimination diet or allergen trial under medical guidance. The goal for this is to make it so that the users could log in and track their allergies from food.

Product Architecture

Sections and Routes

- **Dashboard (Home):** Overview of active trials, recent symptoms, and quick access to start a new trial.
- **Trial Tracker (Active Trial):** Detailed view of an ongoing food trial where users log symptoms, severity, and ultimately flag the food as safe or unsafe.
- **Food Log (History):** A searchable directory of all completed trials and their safety status
- **Allergen Scanner:** A camera/image upload interface to picture food and determine if it contains any previously marked allergens through gemini integration.

State Management

The application manages three primary data layers to ensure real-time tracking and historical accuracy.

Data Object	Properties	Owner	Trigger
trials	Array of { id, foodName, startDate, status, symptoms: [{date, severity, notes}] }	App Context	User adds trial, logs symptom, or update status
currentTrialID	String (ID)	Router/App	User navigates to a specific active trial from the dashboard
newSymptomInput	Object { severity: number, notes: string }	Symptom Form Component	User adjusts severity slider or types in notes

Screen Breakdown: Trial Tracker (Active Trial)

The Trial Tracker serves as the core functional interface during the elimination diet process.

- **Block 1: Trial Header**
Displays the current food being tested, the start date, and a counter of total days elapsed to ensure the trial meets the required duration.
- **Block 2: Symptom Logging Form**
The primary interaction point. It contains a severity scale (1-10), a text input for symptom notes, and a "Log Reaction" button for quick entries.
- **Block 3: Symptom Timeline**
A chronological list of all symptoms logged for this specific food trial, allowing users to see patterns over the course of several days.
- **Block 4: Verdict Controls**
High-emphasis buttons used to finalize the trial by marking the food as "Flag as Safe" or "Flag as Unsafe."

Content Requirements

- **Sample Trial Data:** Mock food entries, dates, and symptom history to build and test the interface responsiveness.
- **Medical Standard Scales:** Symptom severity scales or criteria provided by to ensure the logging is medically useful and standardized.
- **API Research:** Documentation and access for an image recognition/food analysis API (e.g., Clarifai, Google Cloud Vision, or a specific food database API) for the Allergen Scanner feature.

Technical Risk Assessment

The "Picture Food to Determine Allergens" Feature

Implementing image recognition for complex dishes is difficult and heavily relies on external APIs or AI models. It might be challenging to accurately guarantee a specific allergen is present just from a picture. This carries significant safety risks for a medical-use app, as false negatives could lead to severe allergic reactions.